

mid exam - 2023

Correct answers

1	C	6	B	11	B	16	B	21	B
2	C	7	A	12	B	17	C	22	C
3	B	8	C	13	A	18	D	23	B
4	C	9	C	14	A	19	B	24	D
5	D	10	B	15	C	20	C	25	D



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- 1- Each atom in a silicon crystal has:
- ☒ a- four valence electrons
 - ☐ b- four conduction electrons
 - ☒ c- eight valence electrons, four of its own and four shared
 - ☐ d- no valence electrons because all are shared with other atoms

2- When a voltmeter is placed across a forward-biased diode, it will read a voltage approximately equal to:

- ☐ a- the bias battery voltage
- ☒ b- 0 V
- ☐ c- the diode barrier potential
- ☐ d- the total circuit voltage

* For forward bias for the silicon diode given $I_D = 4 \text{ mA}$ at 0.58 V answer the question (3 and 4)

3- the ac resistance at given current equal

- ☐ a- 6.5Ω
- ☒ b- 13Ω
- ☐ c- 145Ω
- ☐ d- 290Ω

4- The dc resistance at this current level is

- ☐ a- 6.5Ω
- ☐ b- 13Ω
- ☒ c- 145Ω
- ☐ d- 290Ω

5- The depletion region consists of

- ☐ a- nothing but minority carrier's
- ☐ b- positive and negative ions
- ☐ c- No majority carrier's
- ☒ d- answer (b) and (c)
- ☐ e- Neither (a), (b), nor (c)

6- If you are checking a 50 Hz full-wave bridge rectifier and observe that the output has a 50 Hz ripple

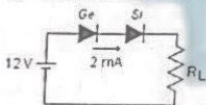
- ☐ a- the circuit is working properly
- ☒ b- there is an open diode
- ☐ c- the transformer secondary is shorted
- ☐ d- the filter capacitor is leakage

7- Determine the voltage across the resistor $3 \text{ k}\Omega$.



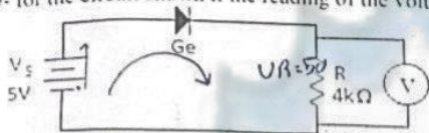
- ☒ a- 0 V
- ☐ b- 0.9V
- ☐ c- 0.2V
- ☐ d- 0.44V

8- Determine the value of the load resistor.



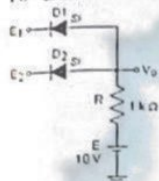
- ☐ a- $5 \text{ k}\Omega$
- ☐ b- $6 \text{ k}\Omega$
- ☒ c- $5.5 \text{ k}\Omega$
- ☐ d- $11 \text{ k}\Omega$

9. for the circuit shown if the reading of the voltmeter is 5 V so the diode is:



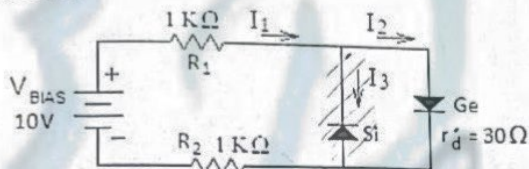
- a- in normal operation ☒ b- open c- short d- cannot be determined

10- Determine the current through each diode if $E_1 = E_2 = 0$ V.



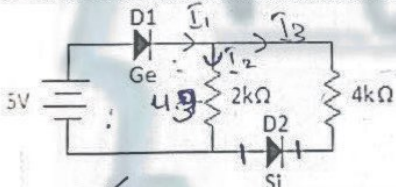
- a-0.7 mA ☒ b-4.65 mA c-9.3 mA d-18.6 mA

for the circuit shown, assume that both diodes are complete piecewise (including r_d). Answer questions 11, 12, 13



- 11- the value of the current I_1 equal
a-0 mA ☒ b-4.78 mA c-5.58 mA d-4.65 mA e- 5 mA
- 12- the value of I_2 equal
a-0 mA ☒ b-4.78 mA c-5.58 mA d-4.65 mA e- 5 mA
- 13- the value of I_3 equal
a-0 mA ☒ b-4.78 mA c-5.58 mA d-4.65 mA e- 5 mA

14- in the circuit below the value of the current through the resistor 4 kΩ equal



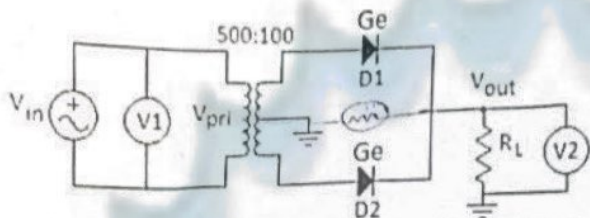
- a-0 mA ☒ b-1.075 mA c-1 mA d-1.25 mA

15- the voltage across D2 in circuit question 14 is

- a-0 V ☒ b-5V ☒ c-4.3 V d-0.7 V

4.7

* For the circuit shown answer the questions 16, 17, 18, 19, 20, 21



$$V_{dc} = \frac{2}{\pi} V_{out}$$

16- If the peak value of the output is ($V_{out} = 2.2$ V) the reading of the voltmeter V_2 is

- a- 2.74 V b- 1.4 V c- 0.68 V d- 3.04 V

17- The maximum value of the secondary voltage (V_{sec}) equal to

- a- 8.6 V b- 15 V c- 5 V d- 10 V

18- The maximum value of the primary voltage (V_{pri}) equal to

- a- 5 V b- 35 V c- 45 V d- 25 V

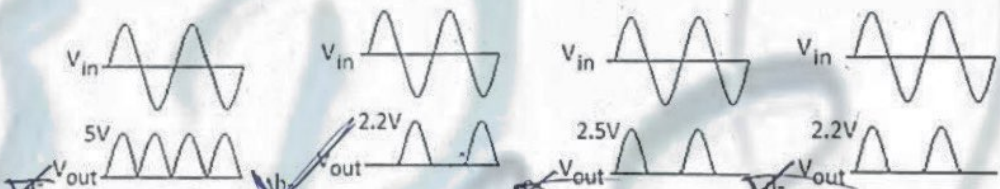
19- The reading of the voltmeter V_1 is

- a- 35.35 V b- 17.68 V c- 42.43 V d- 21.21 V

20- PIV for each diode is

- a- 10 V b- 6.2 V c- 9.3 V d- 4.7 V

21- If in the circuit the diode D_1 is open so the output pulses will be



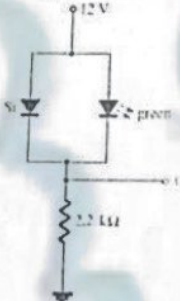
22- for practical (zener diode) in forward bias the voltage across its terminal is

- a- 0 V b- the applied voltage c- 0.7 V d- the Zener diode voltage

23- By addition of pentavalent to the silicon material the hole current is became

- a- Majority carrier b- Minority carrier c- No change the hole current

24- For the circuit shown, if the forward voltage of the green diode is ($V_d = 2$ V) the value of the current through the LED is



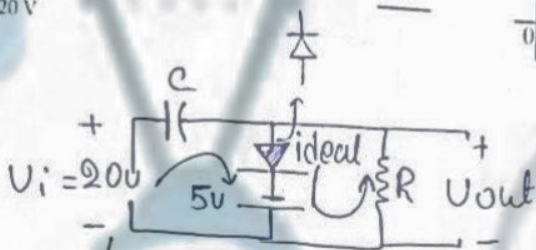
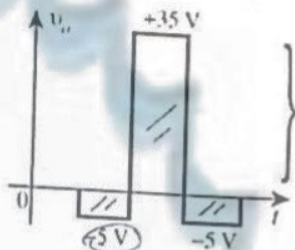
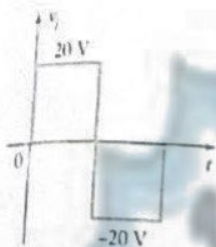
- a- 4.54 mA b- 5.136 mA c- 5.454 d- 0 mA

25- While making the test for the Zener diode circuit the output voltage received less than the Zener voltage in this case the Zener current is

- a- $I_Z = I_{Zmax}$ b- $I_Z = I_R$ c- $I_Z = I_L$ d- $I_Z = 0$ mA

Problem 2

Design a clamper to perform the function indicated in figure shown (the diode in the circuit is ideal)



$v_{out} =$
 @ positive terminal

$$-20 - 5 + V_c = 0$$

$$V_c = 25V$$

$$V_{out} = -20$$

diode $\rightarrow$ short circuit
 forward bias

$$v_{out} - v_i + V_c = 0$$

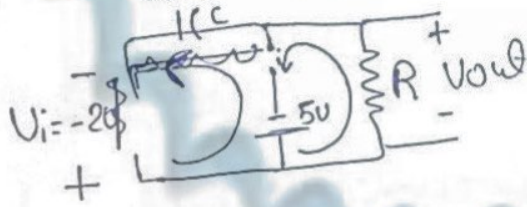
$$v_{out} = v_i - V_c$$

$$20 - 25 = -5V$$

$$R = V_{out} = -5V$$

@ negative terminal
 diode become open circuit @ Reverse Bias.

$$v_{out} = PIV - 5 =$$



$$5 + PIV - V_c - v_i = 0$$

$$PIV = v_i + V_c - 5$$

$$20 + 25 - 5$$

$$PIV = 40V$$

$$v_{out} = PIV - 5 =$$

$$40 - 5 = 35V$$